INCH-POUND

MIL-C-50785/16A (AR)
27 November 1992
SUPERSEDING
MIL-C-50785/16 (MU)
6 October 1975

# MILITARY SPECIFICATION SHEET

## CIRCUIT CARD ASSEMBLY: 12000016

## (RANGE SERIES)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50785.

REQUIREMENTS

1 Item definition. This assembly is a single-ended printed circuit board consisting of three transistors, six operational amplifiers and three divider modules. The assembly is used to generate the range series input term of the magnet current voltage circuitry when installed in the Sight Current Generator Drawing Number 12000239.

2 Characteristics.

3 Performance. The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000016.

a. Operational amplifiers AR1 and AR2 in conjunction with Q1 and Q2 shall generate +7 Vdc reference voltage, which is based on the -10 Vdc radar reference voltage, when the sight current generator is operating in the radar mode and -7 Vdc when the sight current generator is operating in the manual mode.


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b. Operational amplifier AR3 and AR4 in conjunction with Q3 shall amplify and scale the range input voltage.

c. Operational amplifier AR6 shall generate the +7 Vdc reference voltage.

d. Dividers U1, U2 and U3 in conjunction with operational amplifier AR5 shall develop and scale the range correct term.

4.1 Input and output levels.

4.2 Input power. Input power shall be supplied to the assembly as specified in Table I.

TABLE I. Input voltages.

| PIN | VOLTAGES |
|---|---|
| 14 and 15 | +15 Vdc ± 2.0 percent |
| 3 and 4 | -15 Vdc ± 2.0 percent |
| 1 and 2 | Ground |
| 21 | -7 Vdc ± 0.1 percent |
| 5 or 6 | Triangle wave -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 1.00 milliseconds ± 10 percent. |
| 7 or 8 | Triangle wave -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 0.67 millisecond ± 15 percent. |

4.3 Output levels. The requirements specified in 4.4 through 4.7 shall be met when supplied with the specified voltage input. The inputs applied to the microcircuits U1, U2 and U3 shall not exceed the electrical requirements specified on Drawing Number 12000091.

4.4 Pin 9 output. The output between pin 9 and pin one shall be 1.568 (U1) - 1.425 (U2) + 0.730 (U3) ± 4.0 percent with a Vin applied between pin 11 and pin one where:

$$U1 = \frac{4.9}{(-)\ (0.5833)(Vin)}$$

$$U2 = \frac{0.7U1}{(-)\ (0.5833)(Vin)}$$

$$U3 = \frac{0.7U2}{(-)\ (0.5833)(Vin)}$$

Vin = + 1.0 Vdc to +12 Vdc with +15 Vdc applied to pin 18
or, Vin = + 1.25 Vdc to +15 Vdc with - 15 Vdc applied to pin 18



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4.5 Pin 13 output. The output between pin 13 and pin one shall be as specified in a, b and c.

a. Zero ± .006 Vdc with zero Vdc applied between pin 11 and pin one, and -15 Vdc applied between pin 18 and pin one.

b. Vin (+0.4666) ± 1.0 percent with Vin of zero to +15 Vdc applied between pin 11 and pin one, and -15 Vdc applied between pin 18 and pin one.

c. Vin (+0.583) ± 1.0 percent with Vin of zero to +15 Vdc applied between pin 11 and pin one, and +15 Vdc applied between pin 18 and pin one.

4.6 Pin 22 output. The output between pin 22 and pin one shall be as specified in a and b.

a. +7.0 ± 0.020 Vdc with -10 Vdc applied between pin 16 and pin one, -7.0 Vdc applied between pin 21 and pin one and +13 Vdc applied between pin 18 and pin one.

b. +7.0 ± 0.020 Vdc with -7.0 Vdc applied between pin 21 and pin one, and -15 Vdc applied between pin 18 and pin one.

4.7 Pin 23 output. The output between pin 23 and pin one shall be +7.0 Vdc ± 0.1 percent when -7.0 Vdc ± .01 percent is applied between pin 21 and pin one.

4.8 Fault isolation test points. The assembly has the following test points available to facilitate fault isolation: pins 10, 12, 17, 20, 24, 25, TPI-6. These are not output pins.

5 Test equipment. The test equipment listed below shall be used in the performance of the applicable test methods and procedures as specified in 6.

5.1 Power supply (PS1). The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.2 Power supply (PS2). The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.3 Power supply (PS3). The power supply shall be +24 Vdc ± 2.0 percent, with regulation to ± .01 percent, 10 ma or greater.

5.4 Power supply (PS4). The power supply shall be -10 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.5 Power supply (PS5). The power supply shall be -7 ± 0.001 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) or 100 mv peak-to-peak.



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5.6 Power supply 01 (PS6). The power supply shall generate a triangle wave -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 1.00 millisecond ± 10 percent.

5.7 Power supply 02 (PS7). The power supply shall generate a triangle wave -8.50 ± 1.50 Vdc to +8.50 ± 1.50 Vdc within a period of 0.67 millisecond ± 15 percent.

5.8 Digital voltmeter (DVM). The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

6 Methods of inspection.

6.1 Environmental. As specified in paragraph 4.7.1 of MIL-C-50785.

6.2 Performance.

6.3 Test connections. Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 1 using receptacle M24308/1-3 of specification MIL-C-24308/1.

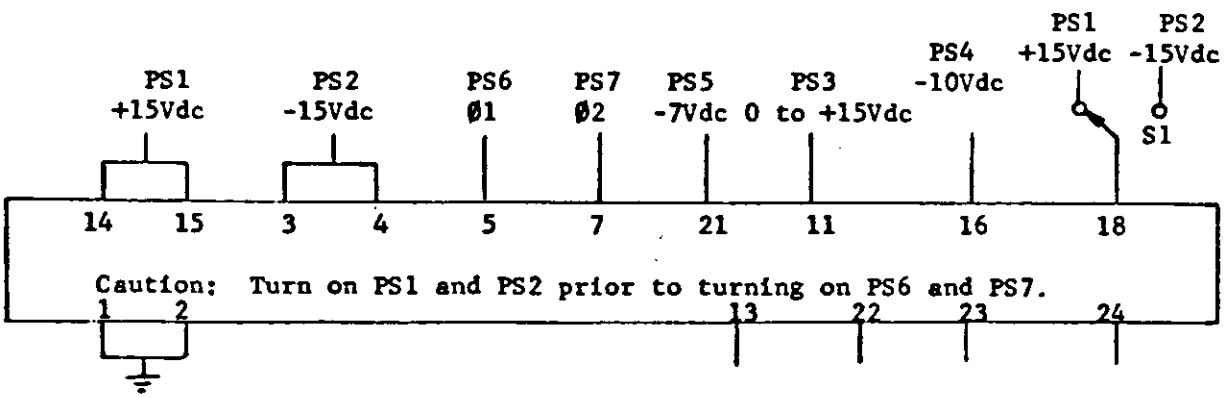



FIGURE 1. Test connections.

6.4 Test procedure. Compliance with the requirements of 4.3 through 4.7 shall be verified as follows:

a. Set switch S1 to +15 Vdc, adjust power supplies PS3 and PS4 as specified in Table II.



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TABLE II. Connections.

| Pin 11 Input Vdc | Pin 16 Input Vdc | Pin 9 Output Vdc |
|---|---|---|
| 1.2 ± 0.007 | -10 ± 0.007 | 6.110 ± 0.245 |
| 4.0 ± 0.007 | -10 ± 0.007 | 2.533 ± 0.100 |
| 8.0 ± 0.007 | -10 ± 0.007 | 1.441 ± 0.058 |
| 12 ± 0.007 | -10 ± 0.007 | 0.997 ± 0.040 |

b. Set switch S1 to -15 Vdc, adjust power supplies PS3 and PS4 as specified in Table III.

TABLE III. Connections.

| Pin 11 Input Vdc | Pin 16 Input Vdc | Pin 9 Output Vdc |
|---|---|---|
| 1.5 ± 0.007 | -10 ± 0.007 | 6.110 ± 0.245 |
| 15 ± 0.007 | -10 ± 0.007 | 0.997 ± 0.025 |

c. Set switch S1 to +15 Vdc, adjust power supply PS3 as specified in Table IV.

TABLE IV. Connections.

| Pin 11 Input Vdc | Pin 13 Output Vdc |
|---|---|
| 0 ± 0.001 | 0 ± 0.006 |
| 12 ± 0.007 | 7 ± 0.070 |

d. Set switch S1 to -15 Vdc, adjust power supply PS3 as specified in Table V.

TABLE V. Connections.

| Pin 11 Input Vdc | Pin 13 Output Vdc |
|---|---|
| 0 ± 0.001 | 0 ± 0.006 |
| 15 ± 0.007 | 7 ± 0.070 |

e. Set switch S1 to +15 Vdc. The output measured at pin 22 with the DVM shall be +7 ± 0.020 Vdc.

f. Set switch S1 to -15 Vdc. The output measured at pin 22 with the DVM shall be +7 ± 0.020 Vdc.

g. The voltage measured at pin 23 with the DVM shall be +7 ± 0.007 Vdc.



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7.0 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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3. DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY: 12000016 (RANGE SERIES)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

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